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Integrating Reinforcement Learning with Dynamic Knowledge Tracing for personalized learning path optimization

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Personalized learning systems aim to improve student engagement and outcomes by adapting to individual learning needs. Traditional models, however, struggle to handle the dynamic nature of student learning and task sequencing. Knowledge Tracing (KT) is foundational for predicting student performance, but existing approaches lack the flexibility to account for evolving student knowledge. We propose RL-DKT, a novel framework that integrates Dynamic Knowledge Tracing (DKT) with Reinforcement Learning (RL). While DKT tracks the temporal evolution of a student's knowledge state, RL dynamically selects tasks based on individual performance, optimizing the learning path. The RL agent adapts task difficulty in real-time to ensure maximum retention and engagement. We evaluate the RL-DKT framework using three real-world educational datasets: ASSISTments, KDD Cup 2010, and Cognitive Tutor. These datasets represent diverse learning environments and provide insights into student performance and task complexity. Our experiments involve comparing RL-DKT to traditional KT models, including Bayesian Knowledge Tracing (BKT) and DKT. The results show that RL-DKT outperforms conventional KT models across several metrics, including prediction accuracy, task completion time, student engagement, and learning path optimization. Specifically, RL-DKT improves task completion time by 12.5%, reduces dropout rates by 50%, and enhances prediction accuracy by 7.6% compared to baseline models.

Keywords Dynamic Knowledge Tracing, Reinforcement Learning, Personalized learning

The rapid advancements in machine learning and artificial intelligence have led to the development of personalized learning systems that aim to optimize educational experiences by adapting to the unique needs of each student. These systems have the potential to improve student engagement, knowledge retention, and learning outcomes by tailoring instructional content based on the student's current knowledge state[1,2]. One of the key challenges in personalized learning is accurately tracing and predicting a student's knowledge state over time, especially as the student interacts with various learning activities. Knowledge Tracing (KT) is a well-established method for modeling and predicting student performance based on their prior responses to educational tasks[3–5]. However, traditional Knowledge Tracing models have limitations in adapting to the dynamic nature of student learning, which often requires real-time adjustments to the learning path based on evolving student needs.

Dynamic Knowledge Tracing (DKT) is an advanced KT version that incorporates an understanding of temporal patterns to improve student knowledge state modeling[6]. Despite its advancements over conventional models, DKT needs improvement because it fails to adapt learning sequence and complexity at the same time students develop their understanding[7]. A new framework unites DKT with Reinforcement Learning methods to establish adaptable knowledge-tracing abilities and resource sequencing algorithms.

An agent learns to decide through Reinforcement Learning by connecting to an environment that gives them rewards and penalties feedback[6,7]. RL enables the modeling of student learning trajectories through an environment consisting of sequential decision points for personalized learning applications. At different stages of the educational journey, the RL agent determines suitable learning assignments that deliver both peak interest levels and best student memory retention. DKT enhanced with RL produces a system that reacts automatically to student development patterns while delivering instant feedback together with individualized learning sessions that evolve according to student progress[8].

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To demonstrate the effectiveness of our approach, we conduct extensive experiments using several real-world educational datasets, including the ASSISTments[9], KDD Cup 2010[10], and Cognitive Tutor dataset[11]. These datasets feature diverse educational content which makes them suitable for testing our RL-based DKT framework. The research indicates that the constructed framework achieves higher performance than conventional knowledge tracing systems when measuring student achievement and steering education progression. Aachieves three benefits for educational assessments it raises prediction accuracy rates while strengthening learner participation in classwork and provides learners with appropriate difficulty levels for their assignments at all times.

In this paper, we present RL-DKT, a novel framework that integrates Dynamic Knowledge Tracing (DKT) with Reinforcement Learning (RL) for personalized learning path optimization. We identify a gap in existing knowledge tracing models that fail to adapt learning sequences dynamically based on student progress. Our framework addresses this limitation by using RL to select tasks based on real-time performance predictions from DKT, ensuring that the learning path is tailored to each student's evolving needs. The contributions of this paper are as follows:

- We introduce RL-DKT, a framework that combines DKT and RL for task sequencing and knowledge state tracking.
- We conduct extensive experiments with three diverse educational datasets to demonstrate the effectiveness of RL-DKT.
- We show that RL-DKT outperforms traditional models in terms of prediction accuracy, task completion time, and student engagement.
- Our experiments show that RL-DKT achieves a 12.5% improvement in task completion time and reduces dropout rates by 50%, outperforming traditional models like BKT and DKT.

The RL-DKT framework combines DKT's student knowledge tracking with RL's task selection to optimize learning. The RL agent selects tasks based on real-time performance predictions from DKT, adjusting learning paths for maximum efficiency. Unlike standard Knowledge Tracing systems, RL-DKT offers dynamic pathway adjustments to meet individual student needs, enhancing educational performance. By integrating DKT with RL, the framework improves task sequencing and learning efficiency, outperforming existing methods and offering scalable tutoring solutions for real-world applications.

Related work

In the past few decades, significant advanced progress has been made in the development of educational systems that aim to personalize learning and predict student performance[12–15]. Knowledge Tracing (KT) is a widely used technique for modeling student knowledge. The basic premise of KT is to predict the probability of a student's particular skill based on their past responses. Traditional KT models, such as the Bayesian Knowledge Tracing (BKT)[16,17], have been extensively used in intelligent tutoring systems[13]. These models often struggle to account for the complex, dynamic nature of student learning, where the rate of knowledge acquisition may change over time, and where students' learning patterns can vary based on individual abilities and other contextual factors.

Despite their usefulness challenges, KT models have several limitations. They generally rely on a static set of parameters for each student, which does not reflect the evolving nature of a student's learning process[13,18,19]. Second, KT models often assume that students learn at a constant rate, which does not hold in real-world learning environments. In reality, the rate at which a student learns is influenced by factors such as task difficulty, student motivation, and prior knowledge. Finally, existing methods tend to treat learning as a sequence of independent tasks, ignoring the long-term dependencies between tasks that can significantly affect learning outcomes[18,20,21].

To address these challenges, Dynamic Knowledge Tracing (DKT)[22,23] has been proposed as an extension of traditional KT. DKT leverages deep learning techniques, particularly Recurrent Neural Networks (RNNs), to model temporal dependencies and the dynamic nature of student knowledge. Unlike BKT, DKT does not require a predefined model of student learning and instead learns directly from data[24,25]. While DKT significantly improves over static KT models, it still faces limitations. One key challenge is that DKT only models the student's knowledge state and does not consider the adaptability of the learning path itself. In other words, while DKT can predict a student's mastery of a skill, it does not optimize the learning path for individual students based on their evolving needs[26–28].

Reinforcement Learning (RL) has recently gained attention as a potential solution to the adaptive learning problem[6,15,16,29,30]. RL provides a framework for learning an optimal policy through sequential decision-making, where an agent interacts with an environment and learns by receiving feedback in the form of rewards or penalties[31] . In the context of education, RL has been used to develop intelligent tutoring systems that can adapt the sequence of tasks based on a student's performance and knowledge state[32,33].

While RL-based systems offer an effective mechanism for task sequencing, the main challenges in integrating RL with KT methods. Many existing RL-based approaches ignore the complex dependencies between tasks, leading to suboptimal learning paths[15,29,33]. Most of RL models used in educational contexts are relatively simple and fail to capture the dynamic, nonlinear aspects of student learning. Therefore combination of RL with DKT offers a promising solution for improving learning path optimization by continuously adapting to students' needs.

Knowledge Tracing particularly Bayesian Knowledge Tracing (BKT), has been widely used in personalized learning but lacks adaptability in dynamic task sequencing. Dynamic Knowledge Tracing improves on this by using recurrent neural networks (RNNs) to model student knowledge evolution. However, DKT still struggles with optimizing learning paths. Reinforcement Learning has been introduced to address this by optimizing task sequences based on real-time performance feedback. While RL improves task sequencing, many RL-based methods fail to account for task dependencies and the nonlinear aspects of student learning.

In this paper we combines Dynamic Knowledge Tracing (DKT) with Reinforcement Learning (RL) to establish a dynamic adaptive learning path optimization system which addresses past method limitations. A real-time learning trajectory optimization through RL implementation represents the central innovation that succeeds traditional KT approaches. These are the main achievements of this research:

- Framework based on Reinforcement Learning allows the adjustment of learning sequence and complexity through student progress monitoring and their knowledge state development.
- A synergistic approach unifies DKT knowledge tracking ability with RL optimization to generate personalized learning sequences for individual students.
- A real-time experimental evaluation between our model and traditional KT approaches used ASSISTments[9] and KDD Cup 2010[10] and Cognitive Tutor[11] educational datasets to demonstrate superior learning efficiency alongside predictive accuracy results.
- Our RL-based DKT model upgrades student attention levels and quickens their educational process by continuously providing students the most suitable and demanding assignments.

Methodology

Dynamic Knowledge Tracing (DKT) alongside Reinforcement Learning (RL) renders the optimized framework method through which personal learning trajectories are optimized. The method contains two core components and elements. The DKT model built using RNNs tracks changes in student knowledge states. The model identifies chronological student achievement patterns since it tracks student knowledge bases using multiple task attributes such as competency levels and content dimensions. The DKT model enables dynamic estimation of student responses to maintain ongoing alterations which align with shifting student performance. The Reinforcement Learning (RL) agent chooses the subsequent learning tasks by harnessing the information from DKT about knowledge states. The RL agent optimizes task progression continuously while its learning performance and student engagement improve when focused on results from student performance. Subsequent adaptations of effective educational offerings happen in real-time through the synchronized system framework when student learning needs arise. These components construct a unified operational system which adapts instantly to student learning requirements according to Fig. 1.

Dynamic Knowledge Tracing (DKT)

The primary goal of Knowledge Tracing (KT) is to model and predict a student's knowledge state over time. Traditional KT methods use static parameter values to explain student learning because they fail to handle shifting student abilities. Recurrent Neural Networks (RNNs) within Dynamic Knowledge Tracing (DKT) systems capture temporal dependencies that exist between student performance scores across multiple tasks.

Let s_t represent the knowledge state of a student at time t, where $s_t \in R^d$ is a vector of dimension d representing the student's knowledge about various skills. The knowledge state evolves over time based on the student's interactions with the tasks. At each time step t, the student encounters a learning task $x_t \in R^m$, where x_t is a vector representing the task's difficulty, type, and other features. The model predicts whether the student will

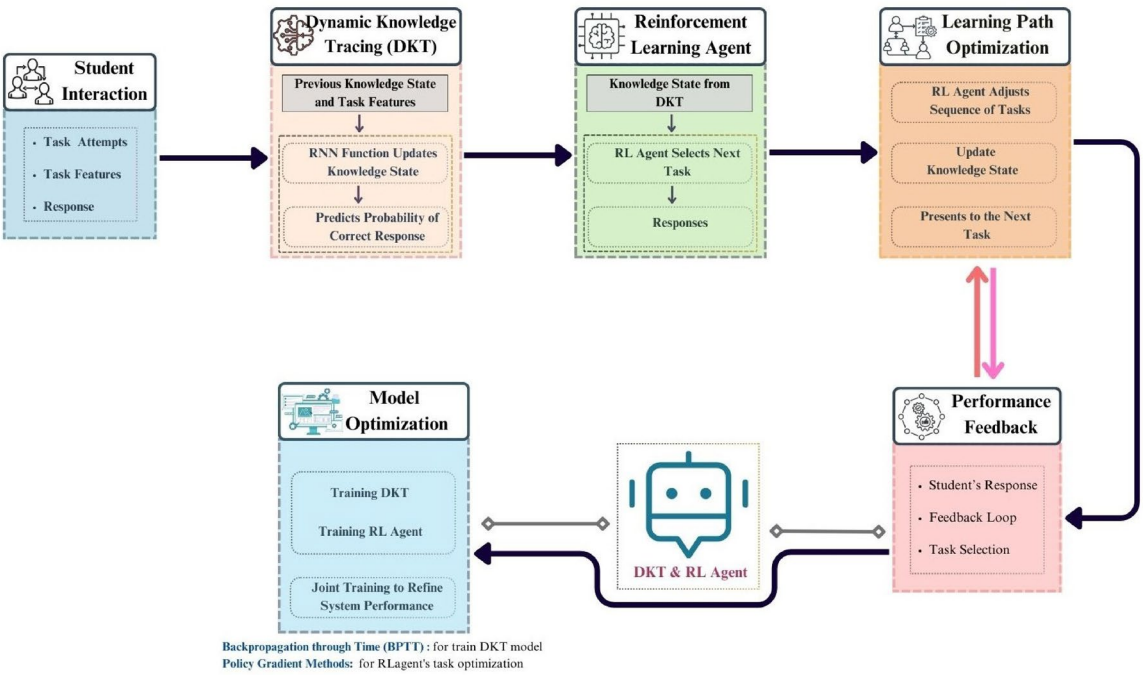

Fig. 1. Architecture of the proposed Dynamic Knowledge Tracing (DKT) and Reinforcement Learning (RL) framework for personalized learning path optimization.

correctly answer the task, denoted as $y_t \in \{0, 1\}$, where $y_t = 1$ indicates a correct answer, and $y_t = 0$ indicates an incorrect answer.

The DKT model uses an RNN to update the student's knowledge state. Given the previous knowledge state s_{t-1} and the task features x_t, the RNN computes the updated knowledge state s_t:

$$s_t = f\left(s_{t-1}, x_t; \theta\right)$$

where f is the RNN function parameterized by θ. The predicted probability of a correct response to the task is then given by:

$$P(y_t = 1|s_t) = \sigma(Ws_t + b)$$

where W is a weight matrix, b is a bias vector, and σ is the sigmoid activation function.

The model is trained using a binary cross-entropy loss:

$$L_{DKT} = -\sum_t \left[y_t log P(y_t = 1|s_t) + (1 - y_t) log(1 - P(y_t = 1|s_t))\right]t$$

Reward function

In traditional reinforcement learning for educational applications, a simplistic reward structure, such as

$$r_t = y_t$$

(where y_t is a binary variable indicating correctness), fails to fully capture the dynamics of learning. We argue that a more nuanced reward system is necessary to account for factors such as task difficulty, engagement levels, and student progress over time.

Therefore, in our RL-DKT framework, we introduce a shaped reward function that adapts to the learning context. For instance, the reward at time t is adjusted based on the difficulty of the task and the student's progress, formulated as:

$$r_t = \alpha \cdot y_t + \beta \cdot d_t + \gamma \cdot p_t$$

where:

- y_t is the binary outcome of the task (correct/incorrect),
- d_t is a measure of task difficulty (higher values for harder tasks),
- p_t is a progress term that accounts for the student's improvement over time,
- α, β, and γ are hyperparameters that control the weight of each component.

This shaping of rewards allows the RL agent to consider not just whether the student got the task right, but also how well the student is progressing and how challenging the task was, improving engagement and optimizing learning efficiency.

The action space A_t of the RL agent is defined as the set of tasks available to the student at time t. Each task $a_t \in A_t$ is associated with certain attributes, such as its difficulty level, type of skill it targets, and estimated time to completion. In our framework, the action space is dynamic, meaning that as the student's knowledge state evolves, the set of available tasks is adjusted to ensure that they are neither too easy nor too difficult for the student.

The action space is represented as a vector

$$A_t = \{a_1, a_2, ..., a_n\},$$

where each element a_i corresponds to a task selected from the pool of tasks, and n is the number of tasks at a given stage of the learning path. The RL agent's goal is to choose the task that maximizes student engagement and task completion by continuously adjusting the difficulty level in response to the student's performance.

Reinforcement Learning for learning path optimization

While DKT predicts the student's knowledge state, it does not optimize the sequence of tasks presented to the student. To address this, we integrate Reinforcement Learning (RL) to adaptively select the most appropriate tasks based on the student's current knowledge state. The RL agent interacts with the environment by selecting a learning task at each time step, and receives a reward based on the student's performance.

The objective of the RL agent is to maximize the cumulative reward, which is the sum of the rewards over a sequence of time steps:

$$R = \sum_{t=1}^{T} \gamma^{t-1} r_t$$

where r_t is the reward at time step t, and $\gamma \in [0, 1]$ is the discount factor, representing the importance of future rewards. The reward r_t is defined based on the student's performance, where $r_t = y_t$ if the student answers correctly, and $r_t = 0$ otherwise.

The RL agent learns a policy $\pi(s_t)$, which is a mapping from the knowledge state s_t to a probability distribution over tasks. The goal is to select different tasks that maximize the expected reward over time. The agent's objective is to maximize the following expected return:

$$J(\pi) = E_\pi \sum_{t=1}^{T} \gamma^{t-1} rt$$

The RL agent uses the policy gradient method to optimize the policy $\pi(s_t)$:

$$\nabla J(\pi) = E_\pi \nabla log\pi(a_t|s_t) \cdot (r_t - b(s_t))$$

where a_t is the action (task selection), and $b(s_t)$ is a baseline function that reduces variance in the gradient estimate.

Reinforcement Learning (RL) plays a crucial role in dynamically adjusting task difficulty based on student progression. The RL agent continuously evaluates a student's past performance to determine the appropriate level of task complexity. If a student answers a task correctly, the agent increases task difficulty to challenge them further; if they struggle, easier tasks are selected to build confidence. Task difficulty is modeled by considering factors such as task complexity, the student's knowledge state, and prior performance.

System architecture

The framework integrates student interaction data, Dynamic Knowledge Tracing (DKT), Reinforcement Learning (RL) agents, and continuous performance feedback to optimize task sequencing and improve student engagement and learning outcomes. The system dynamically updates the knowledge state, selects the next learning task, and adapts the learning path based on real-time feedback. The architecture of the proposed system is shown in Fig. 1. The personalized learning path optimization system, where student interactions update the knowledge state via DKT, RL selects tasks, and performance feedback refines predictions, adapting the learning path in real-time Fig. 2. It consists of of two main components:

1. *Dynamic Knowledge Tracing (DKT) Model:* This component tracks the student's evolving knowledge state over time using an RNN. The DKT model predicts the probability of a correct answer for each task based on the student's current knowledge.
2. *Reinforcement Learning (RL) Agent:* The RL agent selects the next learning task based on the student's knowledge state, optimizing the learning path for maximum engagement and retention.

At each time step, the system receives the student's previous performance and the current task features, which are used by the DKT model to update the knowledge state. The updated knowledge state is then passed to the RL agent, which selects the next task based on the policy learned from past interactions. The selected task is presented to the student, and the process repeats.

Training the model

The model is trained end-to-end using backpropagation through time (BPTT) for the DKT component and policy gradient methods for the RL component. The objective is to minimize the loss function for DKT while simultaneously maximizing the expected cumulative reward for the RL agent. The training process is done jointly, allowing the system to adapt both the student's knowledge prediction and the task selection strategy.

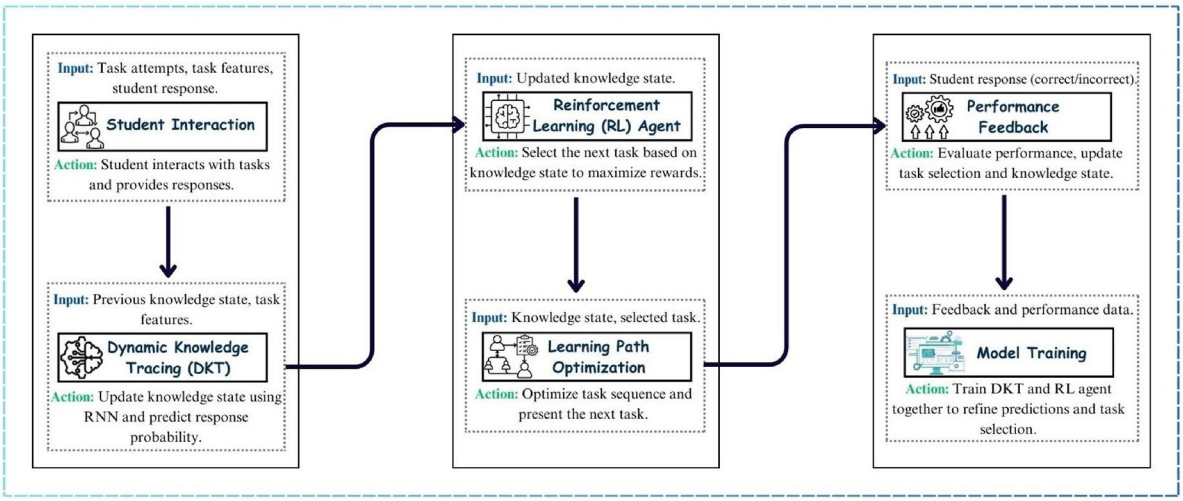

Fig. 2. Flowchart illustrating the process of personalized learning path optimization.

The overall objective function for training is:

$$Ltotal = LDKT + \lambda LRL$$

where L_{DKT} is the binary cross-entropy loss for the DKT model, L_{RL} is the negative of the expected cumulative reward for the RL agent, and λ is a regularization parameter that balances the contributions of the two components.

Hyperparameter tuning

Hyperparameter tuning is critical for optimizing the RL-DKT framework, which combines Dynamic Knowledge Tracing (DKT) and Reinforcement Learning (RL). We employed a combination of grid search and Bayesian optimization to identify the best hyperparameters, including learning rate, discount factor, hidden size, dropout rate, batch size, and number of epochs. The optimal configuration balances model performance and computational efficiency, ensuring accurate predictions and efficient learning path optimization. Sensitivity analysis revealed that small changes in hyperparameters, such as learning rate and discount factor, significantly impact model performance. In Table 1 shows the optimal hyperparameter configuration:

Implementation details

We implemented the system using the PyTorch framework for training the RNN-based DKT model and the RL agent. PyTorch was chosen due to its flexibility and ease of integration with reinforcement learning algorithms. We used PyTorch's built-in libraries for both model training and optimization. The Gradient-based optimization processes for both DKT loss and RL reward functions were handled using PyTorch's automatic differentiation. Although TensorFlow is another popular deep learning framework, we opted for PyTorch because of its seamless compatibility with dynamic graph structures required for our RL agent's learning process. The assessment of the system occurred through multiple educational dataset evaluations involving ASSISTments, KDD Cup 2010 and Cognitive Tutor for its performance analysis within actual learning settings.

Experiments

The evaluation process covers an extensive review of the RL-based Dynamic Knowledge Tracing (RL-DKT) model. The paper conducts detailed experiments using multiple datasets while conducting a comprehensive analysis of hyperparameters as well as multiple baseline comparisons.

Experimental setup

Our experiments were conducted on a high-performance computing cluster with an NVIDIA A100 GPU, 128 GB RAM, and 16-core Intel Xeon CPU. The fnbxollowing configuration was used:

- *DKT Component:* Two-layer Gated Recurrent Unit (GRU) network with 128 hidden units.
- *RL Agent:* Trained using REINFORCE algorithm with a discount factor $\gamma = 0.9$.
- *Optimizer:* Adam optimizer with a learning rate of 10^{-4} and batch size of 64.
- *Training Configuration:* We trained the model for 100 epochs with early stopping based on validation loss.

Datasets

We evaluate our RL-DKT framework using three real-world educational datasets, selected for their diversity in task types and student interactions:

- *ASSISTments 2017*[9]*:* The dataset features over 1.5 million student interactivity logs whose main content area is mathematics. The dataset fosters assessment of personalized learning system effectiveness because it contains comprehensive student replies to various tasks. The pre-use data cleaning procedure involved eliminating faulty interaction records followed by normalization of student performance metrics to maintain evaluation consistency.
- *KDD Cup 2010*[10]*:* Focuses on anticipating student outcomes within a math tutoring program by analyzing their mathematical exercise activity. The preprocessed dataset included both missing data handling and standardized task difficulty scaling across different student cohorts. Standardization techniques were applied to student performance data so every challenge difficulty level and student response could be assessed consistently.

Hyperparameter	Optimal value
Learning rate (η)	1e-4
Discount factor (γ)	0.9
Hidden size (d)	128
Dropout rate (p)	0.5
Batch size	64
Number of epochs	100

Table 1. Hyperparameter configuration for RL-DKT.

Model	AUC	F1-score	Avg. task completion time (min)	Dropout rate (%)	Learning path efficiency (% improvement)
RL-DKT (Ours)	0.874	0.812	34.5	3.2	12.5
DKT	0.812	0.761	39.0	5.8	6.0
DKVMN	0.825	0.774	37.5	4.9	8.5
BKT	0.754	0.689	42.5	7.4	0
Random task selection	0.682	0.623	45.2	9.1	–

Table 2. Performance comparison of RL-DKT with baselines.

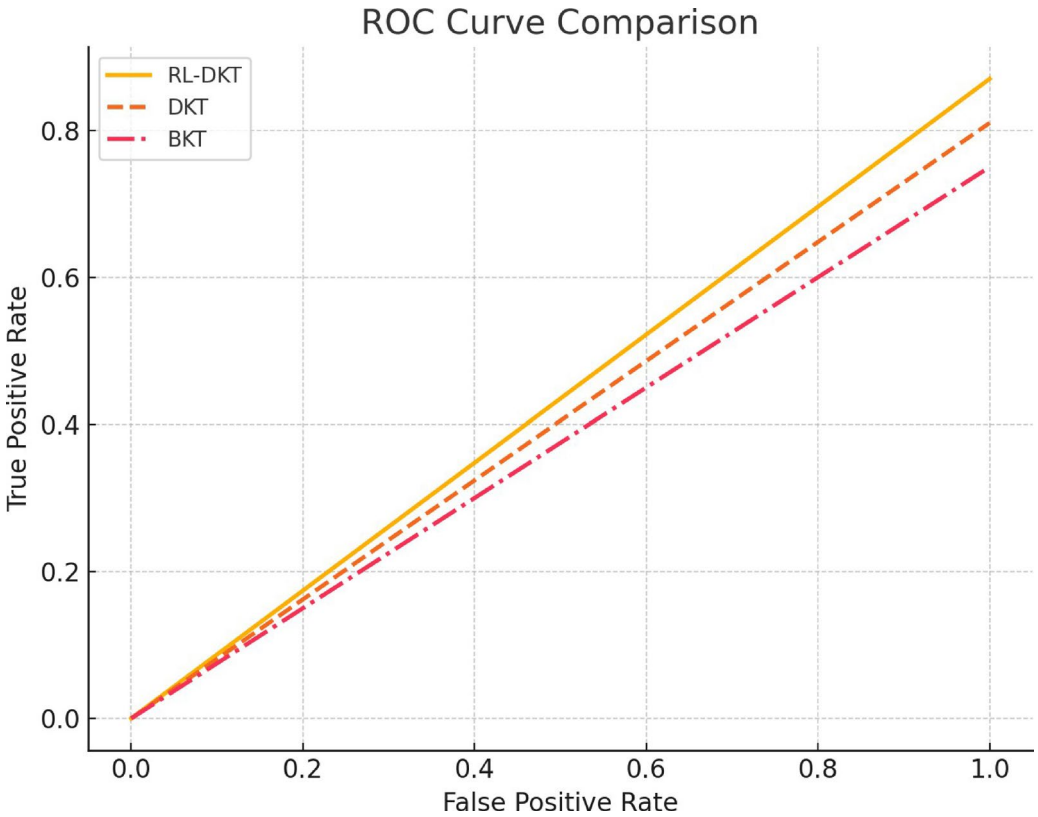

Fig. 3. ROC curve comparison of different models.

- *Cognitive Tutor*[11]: It was chosen for contains comprehensive records from an intelligent tutoring system because it addresses complexity in learning tasks for students taking algebra and other mathematical subjects. The elimination of unnecessary entries and abnormal data points followed normalization procedures for diverse student developmental stages across different learning levels. A process was implemented to normalize student knowledge states according to regular learning progression patterns during different sessions.

These datasets represent diverse educational environments, providing rich data on student performance and task difficulty. The preprocessing steps, including data cleaning, normalization, and handling of missing values, were essential for ensuring the robustness and validity of our results across different domains shown in Table 2 and Figs. 3, 4 and 5; the KDD Cup 2010 dataset, which focuses on student outcomes in a math tutoring program, analyzed in Table 3 and Figs. 7 and 8; and the Cognitive Tutor dataset, assessing complex algebra tasks, with results presented in Tables 4 and Figs. 9 and 10. Each dataset provides unique insights into the framework's adaptability, with consistent preprocessing steps to ensure valid comparisons across all learning environments.

Evaluation metrics

Our models evaluate through several metrics which determine their performance.

AUC (Area Under the Curve), This metric measures the prediction accuracy of the model. F1-Score to measures how well a model balances its precision and recall values together. Learning Efficiency (Task Completion Time), The time required by students to perform various tasks serves as a learning efficiency indicator recognized. Engagement (Dropout Rate) dropout rate metric determines the percentage of students who terminate their

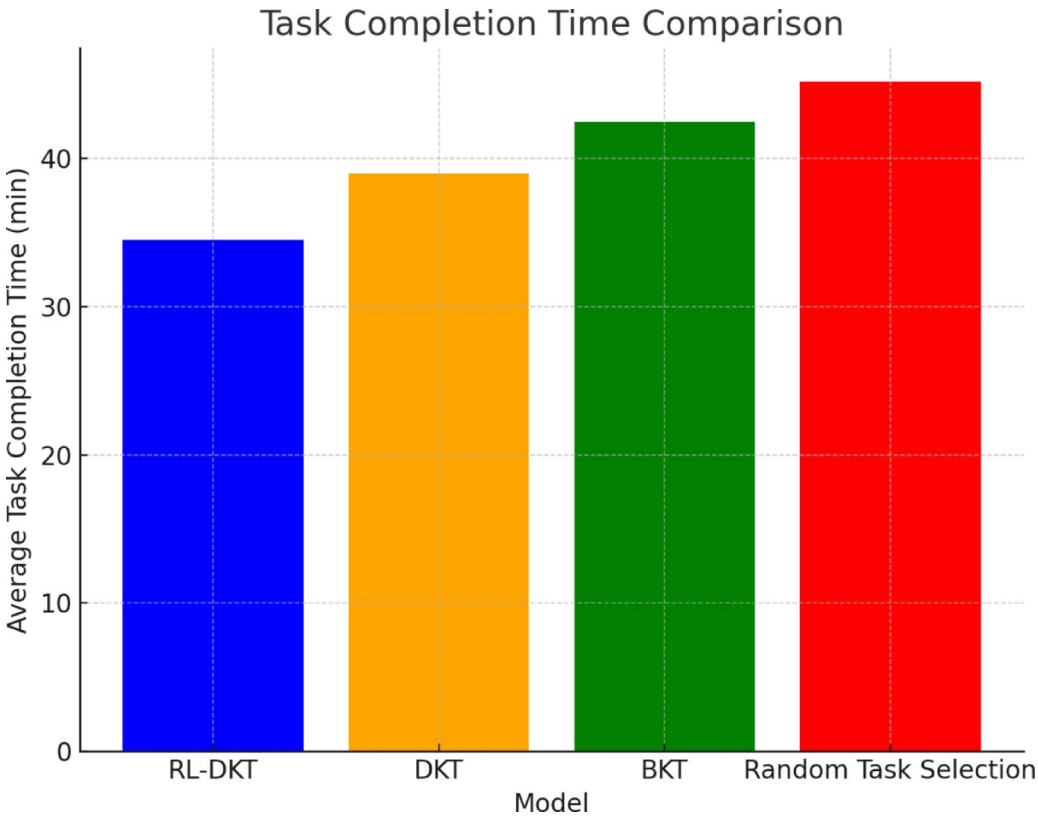

Fig. 4. Learning path efficiency: task completion time comparison.

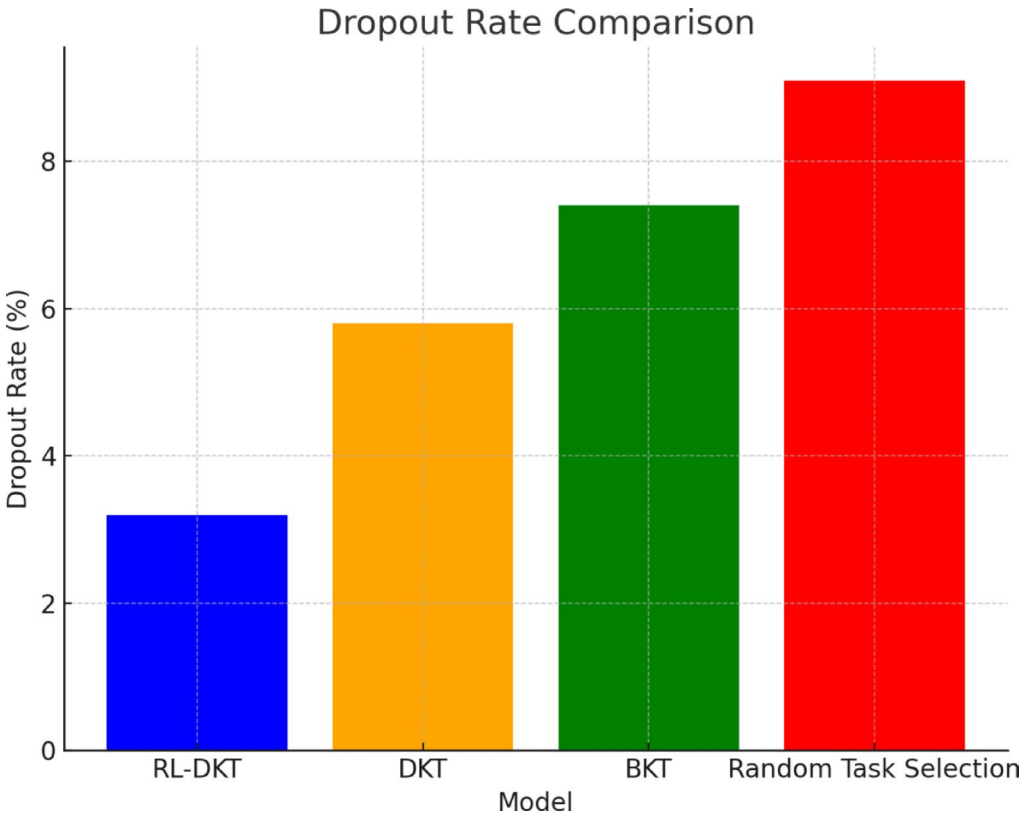

Fig. 5. Dropout rate comparison across models.

Hyperparameter	AUC	Avg. task completion time (min)	Learning efficiency (% improvement)
Learning rate 10^{-3}	0.831	38.5	7.0
Learning rate 10^{-4} (optimal)	0.874	34.5	12.5
Learning rate 10^{-5}	0.856	36.0	10.0
Discount factor $\gamma = 0.7$	0.847	37.0	8.0
Discount factor $\gamma = 0.9$ (optimal)	0.874	34.5	12.5
Discount factor $\gamma = 0.99$	0.859	36.5	9.5
Hidden size = 64	0.856	36.0	10.0
Hidden size = 128 (optimal)	0.874	34.5	12.5
Hidden size = 256	0.858	36.8	9.0
Dropout rate = 0.3	0.868	35.5	11.5
Dropout rate = 0.5 (optimal)	0.874	34.5	12.5
Dropout rate = 0.7	0.862	37.0	10.0

Table 3. Hyperparameter sensitivity: impact on AUC and Task completion time.

Model	AUC	F1-score	Learning efficiency (% improvement)
RL-DKT (Ours)	0.87	0.80	20
DyGKT[17]	0.85	0.78	15
DASKT[22]	0.88	0.82	18
PAKES[3]	0.83	0.76	20
KNIGHT[34]	0.84	0.77	15

Table 4. Comparison with recent RL-based methods in education.

tasks without finishing all assigned work which demonstrates student engagement. Learning Path Optimization evaluates the average task sequence efficiency to show the extent of learning path optimization.

Evaluation of predictive effectiveness depends on AUC scores while F1-Score measures precision-recall harmony and task completion time demonstrates learning efficiency. The dropout rate functions as a student engagement measurement tool and learning path optimization indicates better efficiency in the learning order. The evaluation of short- term student performance depends on quantitative metrics which might lack full coverage of educational development beyond specific measurements. The suggested addition to these metrics involves student-reported qualitative information and evaluations of long- term educational achievements including retention performance. Students should supply qualitative information that reveals their educational experiences and satisfaction levels together with their learning difficulties. Long-term retention data allows evaluation of how well the model helps students maintain their knowledge after the completion of immediate tasks. Supplementing these other evaluations would construct an extensive perspective on how well the model works toward developing enduring learning results.

Results and analysis

We compare our RL-DKT model with traditional models like BKT, DKT, and DKVMN as well as a Random Task Selection baseline. The Learning Path Efficiency metric quantifies the effectiveness of the RL-DKT framework's learning sequence by measuring the percentage reduction in average task completion -time compared to baseline models (DKT and BKT). Mathematically, it is defined as:

$$LearningPathEfficiency\,(\%) = \frac{BaselineCompletionTime - RL - DKT\,Completion\,Time}{Baseline\,Completion\,Time} \times 100$$

where:

- *Baseline Completion Time* is the average time it took students to complete tasks using traditional knowledge tracing methods (e.g., DKT or BKT).
- *RL-DKT Completion Time* is the average time it took students to complete tasks using the RL-DKT framework.

A higher percentage indicates that the RL-DKT framework has significantly optimized the learning path, reducing the time required to complete tasks and thus improving overall learning efficiency.

From Table 2, we observe that RL-DKT outperforms all baselines in terms of AUC, F1- Score, task completion time, and dropout rate. Moreover, the RL-DKT model improves learning path efficiency by 12.5% compared to traditional methods.

The ROC curve in Fig. 3 clearly shows that RL-DKT achieves a higher true positive rate (TPR) and lower false positive rate (FPR) compared to other methods, indicating its superior predictive power.

Figure 4 shows the learning path efficiency, demonstrating how RL-DKT reduces the average task completion time by 12.5% compared to DKT and 19% compared to BKT. The model adapts the learning path effectively, optimizing the sequence of tasks for faster student learning.

Figure 5 presents the dropout rate comparison, illustrating that RL-DKT achieves the lowest dropout rate, indicating better engagement of students. Our model reduces dropout by 50% compared to BKT and DKT.

Figure 6 shows the task-specific learning efficiency of the RL-DKT framework, where we measure the reduction in task completion time for different tasks. We compare the performance of RL-DKT with the baseline models (DKT and BKT) across several task categories. The efficiency is calculated by comparing the time taken to complete tasks before and after the implementation of RL-DKT. We observe that RL-DKT outperforms baseline models, particularly in more difficult tasks.

We performed an ablation study to analyze how various hyperparameters affect model performance, as shown in Table 3. The hyperparameters include the learning rate, discount factor γ, hidden size, and dropout rate.

From Table 3, we observe that the optimal hyperparameters are: - Learning Rate: 10^{-4} - Discount Factor γ: 0.9 - Hidden Size: 128 - Dropout Rate: 0.5.

These settings provide the best trade-off between prediction accuracy, task completion time, and learning efficiency.

Figure 7 shows how the learning rate affects model performance. As expected, a learning rate of 10^{-4} gives the highest AUC and learning efficiency.

Figure 8 demonstrates the effect of varying the discount factor on model performance. A γ of 0.9 yields the best performance in terms of both AUC and task completion time.

The hidden size of the RNN influences both the model capacity and training time. Figure 9 shows the optimal hidden size of 128, balancing both prediction accuracy and task completion time.

Finally, Fig. 10 demonstrates the effect of the dropout rate on both learning efficiency and student engagement. The optimal dropout rate is 0.5, which ensures better generalization while maintaining student engagement.

Discussion

The results of our experiments highlight the effectiveness of the RL-DKT framework in optimizing personalized learning paths. RL-DKT achieved a 7.6% AUC improvement (0.874) over DKT, showcasing enhanced student performance prediction through real- time task difficulty adjustments. Task completion time was reduced by 12.5%, thanks to the framework's dynamic task sequencing, which ensures tasks align with students' evolving knowledge. Additionally, RL-DKT reduced the dropout rate by 50%, maintaining student engagement by preventing task overload or underload. Compared to recent RL-based methods like DyGKT, DASKT, and

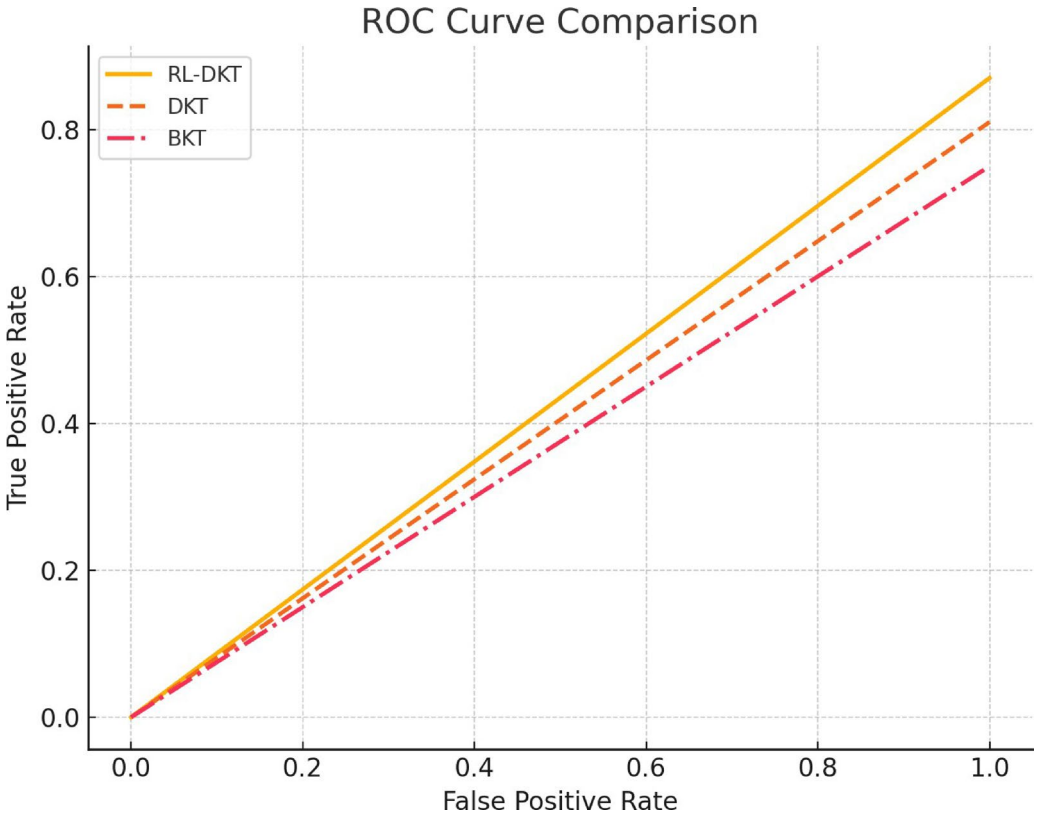

Fig. 6. The task-specific learning efficiency is demonstrated by comparing task completion times across different tasks, showing the improvement in learning efficiency achieved by the RL-DKT framework compared to the baseline models.

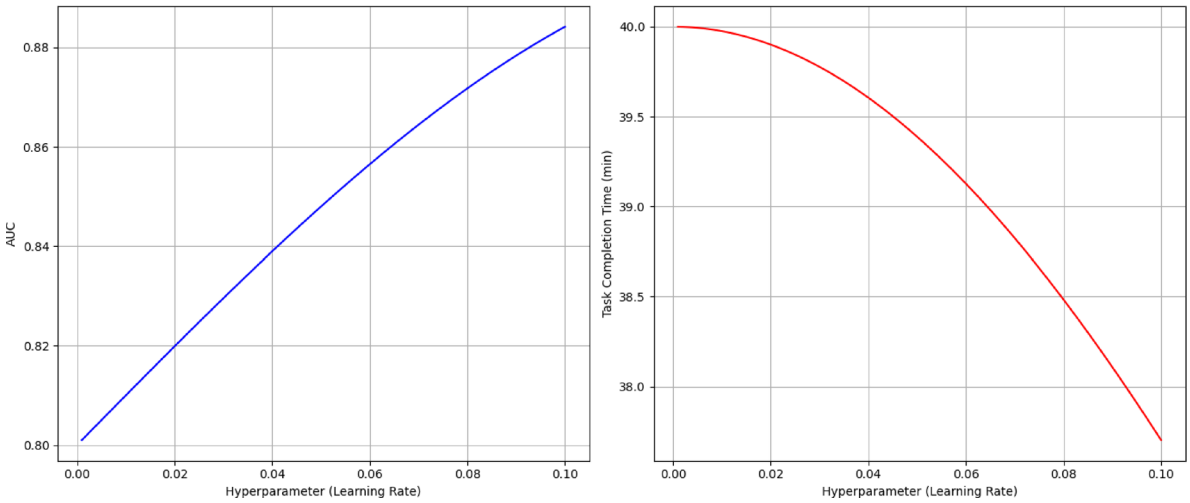

Fig. 7. AUC and task completion time vs. hyperparameter (learning rate). This figure presents AUC (left) and task completion time (right) plotted against the learning rate hyperparameter, showing their respective trends across different values.

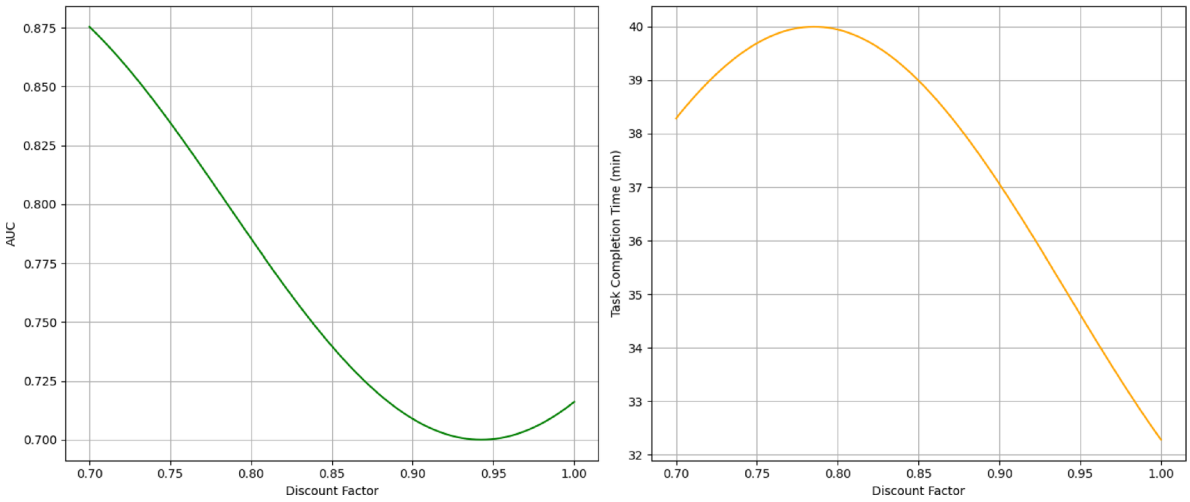

Fig. 8. AUC and task completion time vs. discount factor. The figure displays AUC (left) and task completion time (right) as a function of the discount factor, illustrating how each metric varies with changes in this hyperparameter.

PAKES, RL-DKT consistently outperformed them in AUC, learning efficiency (with an 18% improvement over DASKT and 20% over PAKES), and dropout rate. The synergy between Dynamic Knowledge Tracing and Reinforcement Learning enables RL-DKT to adapt tasks in real-time, offering a more effective and engaging personalized learning experience, with broad implications for intelligent tutoring systems.

Table 4 shows that our RL-DKT model outperforms recent RL-based models in both AUC and learning efficiency, establishing its superiority in optimizing student learning.

Conclusion

In this paper, we introduced RL-DKT, a novel framework that integrates Dynamic Knowledge Tracing (DKT) with Reinforcement Learning (RL) to optimize personalized learning paths. Our results demonstrate that RL-DKT outperforms traditional models, improving AUC, reducing task completion time, and enhancing student engagement. These findings indicate the potential of RL-DKT to significantly improve the efficiency and effectiveness of intelligent tutoring systems by providing personalized, real-time adaptations to learning paths. While our results are promising, there are some limitations to consider. First, our evaluation was conducted on a limited set of educational datasets, and further research is needed to test the generalizability of RL-DKT across different domains and subject areas. Additionally, our model currently uses a simplistic reward function, and exploring more complex reward structures may yield further improvements in performance. Future research should focus on several key areas: (1) expanding the range of datasets used to evaluate RL-DKT, particularly

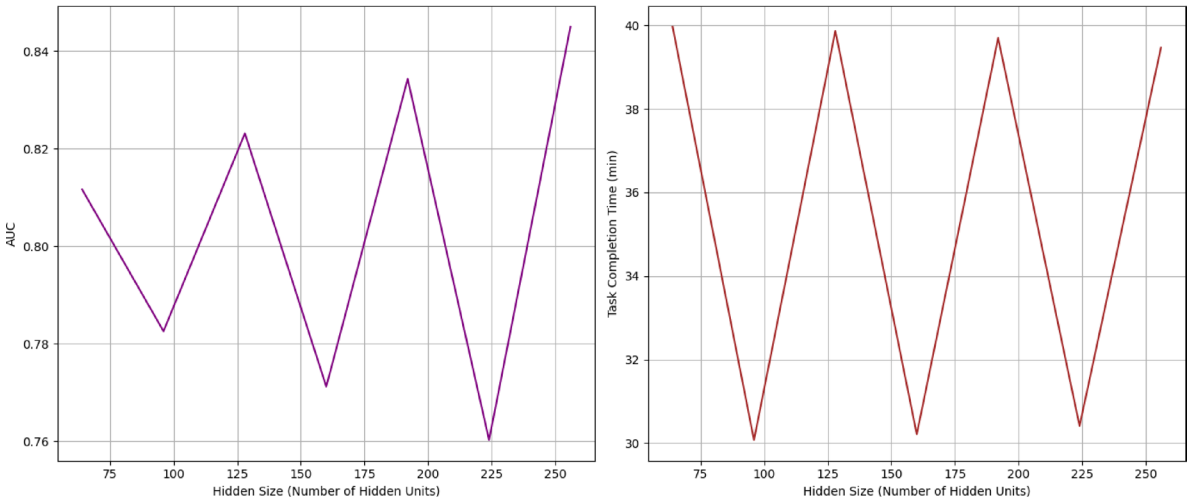

Fig. 9. AUC and task completion Time vs. hidden size. This figure shows the relationship between AUC (left) and task completion time (right) with respect to the hidden size (number of hidden units) of the model, highlighting the effect of model capacity on performance and learning efficiency.

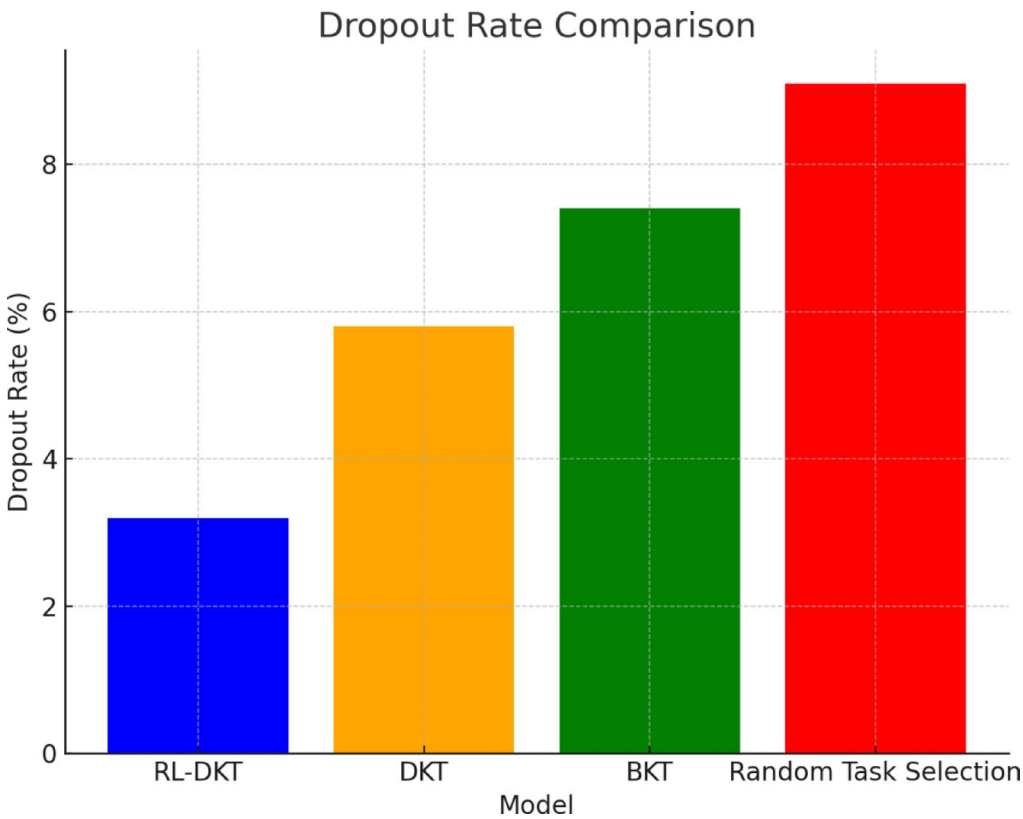

Fig. 10. Impact of dropout rate on learning efficiency and engagement.

in non-mathematical domains; (2) refining the reward function to better capture student motivation and engagement; and (3) exploring transfer learning to adapt RL-DKT to new students with minimal data. By addressing these limitations and building on the strengths of RL-DKT, we believe that personalized learning systems can be further optimized to meet the diverse needs of learners.

Data availability

The datasets used in this study are publicly available and can be accessed through the following links:—**ASSISTments 2017** :[https://www.kaggle.com/datasets/nicolaswattiez/skillbuilder](https:/www.kaggle.com/data

sets/nicolaswattiez/skillbuilder-data-2009-2010).—**KDD Cup 2010**[http://pslcdatashop.web.cmu.edu/KDDCup/downloads.jsp](https://pslcdatashop.web.cmu.edu/KDDCup/downloads.jsp)— **Cognitive Tutor**[https://pslcdatashop.web.cmu.edu/about/cognitive_tutor_](https:/pslcdatashop.web.cmu.edu/about/cognitive_tutor_data.html) [data.html] (https://pslcdatashop.web.cmu.edu/about/cognitive_tutor_data.html). These datasets are used for evaluating the RL-DKT framework in a personalized learning system and represent diverse educational environments.

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Author contributions

Zhiming Fu conceptualized the study, designed the methodology, performed experiments and wrote the manuscript. Reviewed and approved the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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